

ACTINOMYCOSIS

Epidemiology

Described by Israel in 1878, actinomycosis has a worldwide distribution. It is more commonly seen in males aged 20 to 60 years, while females tend to be affected at a younger age. In the pre-antibiotic era, the incidence in the Netherlands and Germany was approximately 1:100,000 inhabitants per year. By the 1970s, the reported incidence in Cleveland, Ohio, had decreased to 1:300,000 per year, while in 1984, in Cologne, Germany, it was estimated at 1:40,000 per year.

Recently, an increased incidence of genitourinary actinomycosis has been observed in females, particularly associated with the use of intrauterine devices (IUDs). Currently, there is a trend toward milder, more localized forms of the disease, with many cases confined to the oral cavity. Diagnosis in these subtle presentations requires a high index of clinical suspicion.

Etiology and Pathogenesis

Actinomycosis is caused by endogenous, anaerobic or microaerophilic, Gram-positive, non-spore-forming bacteria belonging to the families *Actinomycetaceae* and *Propionibacteriaceae* (order: *Actinomycetales*), and *Bifidobacteriaceae* (order: *Bifidobacteriales*). These organisms normally colonize human and animal mucosal surfaces, including the oral cavity, upper respiratory tract, gastrointestinal tract, and female genital tract, with considerable host specificity.

Human pathogens include *Actinomyces israelii*, *A. naeslundii*, *A. gerencseriae*, *A. viscosus*, *A. odontolyticus*, *A. meyeri*, as well as *Propionibacterium propionicum* and *Bifidobacterium dentium*.

In most cases, actinomycosis is a polymicrobial infection involving other commensal organisms from the same mucosal sites. Thus, it is considered a

synergistic infection, with *Actinomyces* acting as the principal ("guiding") organism that determines the disease's course, symptoms, and prognosis. The accompanying flora may include one to nine different species, such as coagulase-negative staphylococci, *Staphylococcus aureus*, α - and β -hemolytic streptococci, microaerophilic and anaerobic streptococci, *Fusobacterium*, *Bacteroides* spp., and other *Propionibacterium* species.

The associated microbiota influences the clinical presentation. Lesions involving pyogenic bacteria like *S. aureus* or β -hemolytic streptococci tend to be acutely inflammatory and painful. In contrast, when anaerobes predominate, the disease follows a subacute, insidious course. A particularly chronic form is observed when *A. israelii* or *A. gerencseriae* is accompanied by *Actinobacillus actinomycetemcomitans*.

Improved culture techniques in recent years have enabled the isolation of multiple *Actinomyces* species from single lesions, suggesting that earlier reports of single-pathogen involvement were likely due to technical limitations.

Infection typically begins at a site of mucosal disruption, which serves as the portal of entry for the organisms.